

Muon Tomography and its Applications.

Rylen Lempa, Dr. Michael Eads

Northern Illinois University

Abstract

Muon tomography or muography utilizes the muon particles produced by the cosmic rays colliding with Earth's atmosphere to digitize the Earth's surface or analyze structures. This method proves to be noninvasive, efficient, and overall, more accurate than other methods. These muons pass through light matter easily, making them useful tools to identify more dense structures or mineral deposits. They also prove useful to more modern global safety applications such as screening for radioactive materials and identifying active volcanoes. Although we have only just begun to explore their potential, there are many other areas of research that these muons could be considered, such as civil engineering to visualize the interiors of historical or sacred buildings. This method is relatively new and has only just begun to reach its potential, but it could have long standing impacts on the way we start to analyze and scan new celestial bodies without leaving our massive footprint like we have on the Earth.

What is Muon Tomography?

Muon tomography or muography, as it will be referenced as for the rest of this article, is a tomographic method which uses muon particles to create an image map of various objects or structures. Muography is a relatively new technique, with the earliest attempts at imaging structures going back to 1970 when Luis Alvarez used cosmic ray muons to try and detect hidden rooms within the Pyramid of Giza [1]. Muography has a

multitude of benefits to it compared to other tomographic methods, some of which include the non-invasive nature of muons themselves. They are microscopic particles that pass through most matter and only interact with the densest materials, making them a vital tool for detecting specific elements or analyzing structures without interacting with said structures. These features have made muography an attractive option in the civil engineering field, as it provides a method that is accurate, cheap, and safe. (BE SURE TO PROVE THESE THREE POINTS IN THE PAPER)

Muons

Before we can understand what, a muon is, we must understand where they come from. The emptiness of the universe makes up about 80% of the total volume of the known universe, with stars and planets and such, filling the remaining space. And through that empty space, there are cosmic rays traveling near the speed of light, which allows them to survive long journeys across the universe thanks to time dilation. These cosmic rays consist of various particles but the most important ones to us are the pions. Denoted in symbol form as the commonly known pi (π). As these pions rain down from space and into Earth's atmosphere, they will collide with the air molecules, causing them to break apart into muons (μ) and neutrinos (ν). Now we have the particle we want; muons are a fundamental particle of the Lepton group. Leptons are fundamental particles that do not interact with the strong force. This strong force is the force that repels subatomic particles apart. This is the key to why muons only collide with only the densest materials, since they ignore the strong force, they rely on direct collisions to be repelled. Muons are very similar

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to electrons with how they behave, with the main difference being that muons are approximately 200 times heavier than electrons. Now that we understand why these muons behave the way they do, we can now discover ways to detect them. Since these particles will pass through most matter without a trace, we must create a dense material that will not only stop them, but also notify us when a muon is halted. These materials are called scintillators; that will fluoresce when struck by a high energy particle. Since the muons are much heavier than electrons and they travel near the speed of light, they are indeed high energy particles.

Background in Scintillators and Tomography

Scintillators are a material that is used to produce illumination when struck by radiation. For muon tomography, slow moving muons collide with the scintillator and excite the atoms, causing them to undergo rapid de-excitation, which releases low energy photons. These flashes of light are then used in tandem with photomultipliers to produce a tangible result to be analyzed. There are two types of scintillators based on the material used, organic or inorganic. Muon tomography uses organic scintillators since this allows a single muon to produce fluorescence. Tomography is a technique that creates a 3-dimensional image of an object from data obtained through various methods. In our case, this data would be obtained from the scintillators and photo multipliers. The first tomographic system was the X-ray CT and went on to provide many applications and led to many other breakthroughs in tomographic methods [2]. The word tomography is derived from Greek, meaning slice or section, as tomography creates slices of an object and then

stacks the slices together to create a 3-dimensional image. There are numerous types of tomography used in all fields of research, among them are electron tomography, ocean acoustic tomography or sonar imaging, quantum tomography, seismic tomography, and x-ray tomography. Muons are particularly unique for tomography since they pass through most matter at high energies, they can be useful for imaging structures or landscapes that are sensitive to regular geographic techniques. Their non-invasive nature is the key component as to why muon tomography stands above other tomographical techniques in certain situations.

Earliest Recorded use of Muon tomography

As stated before, the recorded usage of muon tomography was in 1970 when Luis Alvarez used the technique to scan for hidden rooms inside the pyramid of Giza [1]. According to Alvarez, in the early 19th century, there were several attempts to survey the internal structures of the pyramids, many of which included explosives. The central pyramid had many inner chambers and was more complex than the others, which circulated ideas that the other pyramids hid inner chambers as well. After explosive methods were prohibited for obvious reasons, in the mid-20th century there were proposals to survey the interior of the pyramids using cosmic ray particles. At this time, the properties of cosmic ray particles were well known, having been discovered in 1912 by Austrian physicist Victor Hess [3], but they were not yet utilized for techniques such as this until digital readout devices were created. Alvarez's paper mentions an Australian experiment that first utilized cosmic rays to determine the thickness of ice over an underground tunnel,

but there is little record of this experiment providing any useful results. Initial data from the pyramid experiment was recorded on magnetic tape, with over a million muons being detected over the course of a year. The scintillators were placed in a chamber within the pyramid called the Belzoni chamber near the base, to map out the region of space above. To test this new technique, Alvarez and his team had to demonstrate the validity of the experiment on a known system. However, the only new feature of this technique that required testing was “the magnitude of the scattering of high energy muons in solid matter”. If the scattering is too large, the high resolution of the detectors would prove useless. To test this, Alvarez designated 4 “corners” that were separated by the greatest possible thickness of limestone. This will allow the greatest disparity between corners and create the most opportunity for discovery.

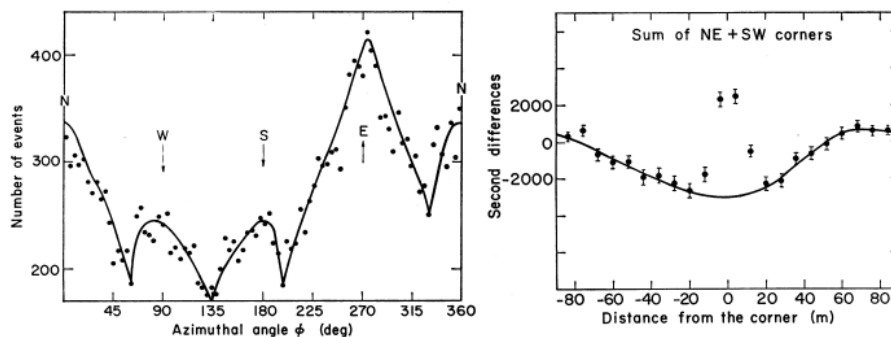


Fig. 3 (left). The initial measurement (with zenith angle of counts from 20 to 40 degrees) of the variation of cosmic-ray intensity with azimuthal angle, as observed from the Belzoni Chamber underneath the Second Pyramid of Chephren. Fig. 4 (right). Detection of the northeast and southwest corners of the pyramid obtained by plotting the second differences of the counting rate on the planes tangent to the corners as a function of distance from the corners.

These figures are pulled directly from Alvarez’s paper, with Fig. 4 being the more interesting of the two. As described by Alvarez, “here the ‘second differences’ of the counting rate with

distance from each corner are plotted on planes that are located symmetrically with respect to adjacent faces and that are tangent to the corner". The results from Alvarez's experiment showed that their initial calculation of scattering turned out to be slightly larger than the measured result. The maximum counting rate is indicated by fig. 4 and Alvarez states that the max counting rate was unexpected; this is due to the detector not lying exactly within the center of the pyramid, so the thickness of limestone varies depending on the direction. The bulk of the paper explains the various mathematical methods and simulations used to visualize the data, but this paper will just focus on overall methods and results. After supplying their simulation with more and more geometric data, slowly mapping out a clearer image, their results were impressively accurate. Another chamber called "The King's chamber" was situated above the Belzoni chamber as well as displaced south and west, directly under the apex. This chamber was used as a reference as there should have been more muons that arrived from that angle through the chamber. It was then simulated again in a scenario where the King's chamber lies just 40 meters above the Belzoni chamber to clearly show the results.

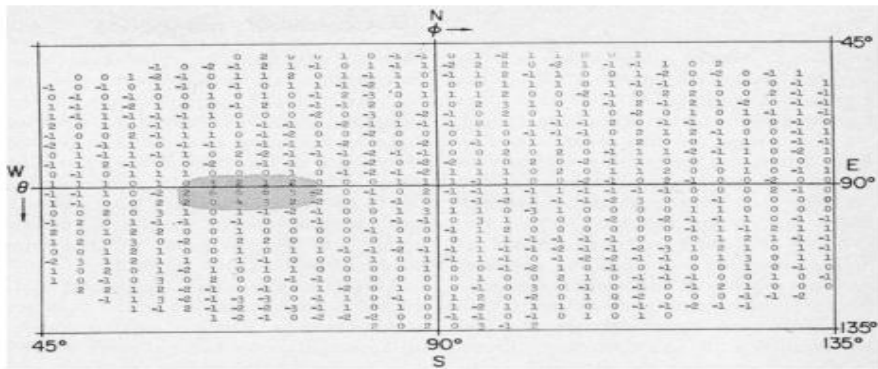


Fig. 11. The display of Fig. 10 as it would have appeared had there been a “King’s Chamber” in the pyramid 40 meters above the apparatus. The group of numbers larger than 3 at the center-left (shaded area) indicates the chamber’s position.

Although using new detectors at the time, the technique of using muons to measure density of objects was not a new idea, still, this experiment remains the origin of modern-day muon tomography due to the undeniable results. Alvarez confirmed that there were no hidden chambers (of similar size and shape of the known chambers) within the Second Pyramid. Muon tomography in this case was proved more accurate, since it covered the entirety of the pyramid, as well as safer than explosive methods, and the only costs were the cost of detectors since the cosmic ray muons are supplied free of charge.

Other Applications of Muon Tomography

The most common use of Muon Tomography in other geosciences is imaging volcanoes, with several dedicated papers to the topic [3]. One such paper titled “High-resolution structural imaging of volcanoes using improved muon tracking”, details the process used to image the density distribution of volcanic edifices [4]. Anytime there is a new algorithm, or a new detector made, there are always researchers eager to test it in the

field. In this case, both a new algorithm and a new detector were being tested at the La Soufrière de Guadeloupe volcano in Lesser Antilles, France [4]. Muon tomography most often combines simulations with raw data such that the data collection does not require long periods of time to be completed. Acquiring muons that decay within a scintillator are far and few between, so simulations are required to fill in the gaps. Using the raw data collected over the course of several months, simulations are supplied with this data to provide more accurate results. This specific method iterates over all muon trajectories to find the best fit, using all the data registered by the detector. Improvements to the data processing reduce the impact that secondary particles have on the data. This leads to higher resolution 2D images and is now able to visualize the asymmetrical structure of the dome. The advantages of using muon tomography to map out these structures as opposed to classic geophysical techniques can be numerous depending on the specific volcano being imaged. The priority being the safety of the workers, is easily achieved since the detectors need only be placed around the volcano and not directly inside potentially dangerous areas. Collection of data is fully automated if the detectors are set up, and the longer the detectors are allowed to collect data, the more accurate the results will be. Another benefit of using muon tomography is that the scale of the structure being imaged can be largely varied, ranging from pyramid chambers that only measure 10 m by 10 m to entire volcano structures and the surrounding regions. Many geophysical techniques are processed with large amounts of noise obscuring the true data, but when you combine the best fit data with a simulation, it will drastically reduce the noise produced, and as the algorithm gains more data to process, the accuracy only improves while the noise

decreases. Of course, muon tomography isn't the sole technique used to fully understand a complex natural structure such as a volcano, it is simply one of several techniques aimed at solving just a piece of the puzzle. Muography is most useful to determine density distribution throughout the volcanic edifices [4]. When used in conjunction with other leading geophysical techniques, the entire puzzle can be completed such that we can predict the behavior of the volcano and mitigate the hazards that may be in the future. 3-dimensional density models can be achieved by combining muon and gravimetry datasets to produce a clear 3D model such that any anomalies would present themselves easier. Not only can muography be useful in preventing disasters before they happen, but they can also be used to recover from natural disasters safely and prevent any secondary disasters from occurring. A recent example comes from the 2011 earthquake Fukushima, Japan [5]. After an earthquake that started a tsunami that destroyed the Fukushima nuclear power plant, many people were worried about a possible nuclear fallout and in 2015, two detectors were set up to scan the site for where the nuclear fuel is located. Hopes were that the nuclear fuel remained in the core of the reactor, but after scanning the site and not detecting any fuel, they had assumed that the fuel either melted down onto the floor or it had breached containment. Muography was used to safely analyze the site without putting any more lives in danger by entering the reactor. In 2017, another muography experiment was conducted on another Great Pyramid by modern archaeologists which uncovered a previously unknown inner chamber at least 30 meters long [5]. This chamber is still yet to be accessed in person, but more research is planned before the chamber is opened. The recent boom in muon tomography is due to all the new technology and devices being

developed. This is the second big push for physics and modern age techniques to be used in this way over the last two decades. After 2001 and the terrorist attacks in New York, there was a substantial effort to improve national security, and along with it came many new techniques that were not tested before. Among them was muography, using large detectors to detect scattered muons to identify dense materials within containers. This idea was simulated by a team of researchers that used two large detectors on either side of the container to determine the muon's path before and after such that any scattering would be prevalent. Not only can the scattering tell you if a dense object is within the container, but it can also lead to exactly what element is within the container, as uranium scatters muons much more than iron does [5]. The scattering method has been shown to be approximately 10 times faster than typical absorption methods with a single detector. This means that for national security purposes, the scattering technique can provide answers within minutes to allow for real time detection. For example, scanning an entire cargo ship for dense materials could take just a few minutes to indicate which shipping containers would need further inspection. In fact, commercial scanners like this are already being utilized in many places around the world to screen cargo ships or even other land borders [5].